HW03 – GUI & Usability Testing Report

Course: Software Testing (CSC13003) – FIT @ HCMUS
Academic Year: 2025–2026
System Under Test: EShop Demo Application (faker26052005-lgtm/SUT-EShop)
Target Repository: `https://github.com/faker26052005-lgtm/SUT-EShop`
Branch: `QuocKy-HW03`
Submission Date: 10/11/2026

Student Information

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1. Executive Summary

1.1 Scope Overview

- **Checklist Screens (Task 1 & 3):** `Cart` (Pool B: FR-07) and `Checkout` (Pool B: FR-08, FR-09).
- **Usability Evaluation Flow (Task 2):** Single End-to-End User Flow: `Product Selection → Add to Cart → Apply Coupon → Checkout`.
- **Target Test Platforms (Task 3):**
    1. **Google Chrome** (Blink engine, Desktop Linux Ubuntu x86_64 / Windows 11)
    2. **Mozilla Firefox** (Gecko engine, Desktop Linux Ubuntu x86_64 / Windows 11)
    3. **Apple Safari / WebKit** (Playwright WebKit / macOS BrowserStack)

1.2 High-Level Results Dashboard

Area	Metric	Target / Requirement	Actual Result	Status
GUI Checklist (Task 1)	Items Designed (IA-01 to IA-04)	> 40 items	**73 items** (70 AI + 3 Human)	**PASSED** (182.5% of target)
	Executed per Platform	100% of designed items	**73 items** per platform	**PASSED** (100% complete)
	Chrome (Blink) Pass / Fail	Audited Pass / Fail	**51 Passed / 22 Failed**	**PASSED** (Audited)

Area	Metric	Target / Requirement	Actual Result	Status
	Firefox (Gecko) Pass / Fail	Audited Pass / Fail	**51 Passed / 22 Failed**	**PASSED** (Audited)
	Safari (WebKit) Pass / Fail	Audited Pass / Fail	**51 Passed / 22 Failed**	**PASSED** (Audited)
Usability (Task 2)	Total Real Participants	Exactly 7 outside class	**7 participants** (`P01`–`P07`) + 1 Pilot	**PASSED** (100% compliant)
	Mean SUS Score	Industry benchmark (~ 68.0)	**76.79 / 100** (Grade B / Good)	**PASSED** (+8.79 vs benchmark)
	Task Completion Rate	Effectiveness metric	**100.0%** (85.7% unassisted)	**PASSED**
	Mean Task Duration	Time-on-task benchmark ≤ 150s	**140.9 seconds** (71.4% ≤ 150s)	**PASSED**
	Friction Points / Findings	Qualitative synthesis	**5 findings** (Ranked by Criticality)	**PASSED**
Cross-Platform (Task 3)	Platforms Covered	≥ 3 distinct platforms	**3 platforms** (Chrome, Firefox, Safari)	**PASSED** (100% coverage)
	Empirical Screenshots	Student overlay on failed items	**81 screenshots** (with email overlay)	**PASSED**
Bug Reporting	Genuine Defects Discovered	Documented with steps & evidence	**24 Bugs** (22 GUI + 2 Usability)	**PASSED**
	GitHub Issues Created	Official issue tracker with labels	**24 Issues** (#117–#138, #159, #160)	**PASSED** (100% verified)
Agent Skills (Task 4)	Reusable Agent Skills	Designed, coded, and demonstrated	**2 skills** (`gui-checklist`, `usability`)	**PASSED**

2. Task 1: GUI Checklist Report (30 Points)

2.1 Scope & Requirement Traceability

Screen	SUT Pool	Functional Requirements Covered	Key Elements & Functional Scope
Cart	Pool B	**FR-07: Shopping cart**	Product items table (thumbnail, title, unit price, quantity, row total, "Xóa" removal button), subtotal summary, "← Mua tiếp" navigation button, "Tiến hành thanh toán" primary CTA button, empty cart state banner.

Screen	SUT Pool	Functional Requirements Covered	Key Elements & Functional Scope
Checkout	Pool B	**FR-08: Checkout FR-09: Discount coupons**	Product summary listing, total payment amount display/input, coupon code input field with "Áp dụng" trigger, promotional feedback banners (success/error/savings), payment method selection (COD), "Xác Nhận Thanh Toán" order submission CTA.

2.2 Taxonomy Grounding & Interface Aspects (IA)

The test suite was designed according to the 4 canonical Interface Aspects (IA) grounded in ISTQB Foundation Level v4.0 §4.1 and Course Lecture S13 (Common GUI Bugs):

1. **IA-01: General UI Standards** (Typography, visual contrast WCAG 2.1 AA, layout alignment, responsiveness, Vietnamese currency formatting `đ` vs `VND`).
2. **IA-02: Forms** (Data validation boundaries EP/BVA, field defaults, required markers, keyboard Tab order, Enter key submission, read-only total integrity).
3. **IA-03: Navigation** (Navbar item count badges, "← Mua tiếp" return links, unauthenticated access route guards, browser Back button state preservation).
4. **IA-04: Feedback / State** (Empty cart notifications, loading spinners, coupon feedback toasts, RAM state vs session persistence upon page reload F5).

2.3 Human-in-the-Loop Review & Gap Analysis

A. AI Item Review & Deduplication

An initial pool of **85 raw checklist items** was generated by Gemini 3.8 Flash. During the Phase 4 systematic review, the student evaluated each item against real SUT architecture, rejecting **19 duplicate, backend-only, or untestable items** to establish a high-value baseline of **70 AI items**.

Total AI Drafts	Kept As-Is	Revised / Refined	Rejected / Merged	Final AI Baseline
85 items	28 items	38 items	19 items	**70 items**

Summary of Rejected AI Items (Audit Defense Log)

Item ID	Screen	IA	Category	Original Check / Intent	Reason for Student Rejection
`CART-IA01-03`	Cart	IA-01	Layout & alignment	Container width 100%	Expected "100% width" contradicts actual responsive design; layout difference between Cart and Checkout is an intentional UX design choice.
`CART-IA01-11`	Cart	IA-01	Labels	Subtotal label capitalization	Redundant with `CART-IA01-10` which comprehensively audits casing consistency.
`CHCK-IA01-02`	Checkout	IA-01	Layout & alignment	Checkout card width	Derived from CSS class (`max-w-lg`) rather than visible requirement; non-functional.

Item ID	Screen	IA	Category	Original Check / Intent	Reason for Student Rejection
CART-IA02-01	Cart	IA-02	Field order & Tab	Tab sequence across Cart	Cart has no text form fields; keyboard navigation was relocated to IA-03 (Focus order).
CHCK-IA02-04..07	Checkout	IA-02	Data validation	Negative, float, overflow, text inputs	Highly fragmented; merged all 5 EP/BVA test vectors into a single comprehensive boundary item CHCK-IA02-03.
CHCK-IA02-09	Checkout	IA-02	Input constraints	Arrow keys / mouse wheel in total input	Merged with CHCK-IA02-08 testing non-editable total integrity.
CHCK-IA02-15	Checkout	IA-02	Control state vs data	Setting total to 0 and submitting	Success screen in Checkout.jsx displays static message without order total; verifying backend DB state is untestable purely via GUI.
CHCK-IA02-23	Checkout	IA-02	Shift+Tab order	Reverse keyboard navigation	Reverse flow integrated directly into CHCK-IA02-22.
CHCK-IA02-25	Checkout	IA-02	Focus on label click	Total amount label focus	Duplicate of CHCK-IA02-24; readonly inputs legally receive focus per HTML standards.
CART-IA04-04	Cart	IA-04	State transitions	Empty to non-empty transition	Reverse of CART-IA04-01 and depends on catalog interactions outside Cart scope.
CHCK-IA04-03	Checkout	IA-04	Error messages	Sub-minimum order error for BIGBUY	Can only be triggered by manually altering total input; invalid user flow.
CHCK-IA04-06..07	Checkout	IA-04	Loading states	Button text "..." and "Đang xử lý..."	Redundant with CHCK-IA02-18 and CHCK-IA02-19.
CHCK-IA04-09	Checkout	IA-04	Currency of data	Page reload (F5) on Checkout	Redundant with CART-IA04-03 (identical RAM state loss mechanism).
CHCK-IA04-11	Checkout	IA-04	State transitions	Form container unmount on success	Redundant with CHCK-IA04-05; DOM unmounting is an implementation detail invisible to users.
CHCK-IA04-15	Checkout	IA-04	Multi-tab sync	Real-time cross-tab synchronization	SUT architecture lacks BroadcastChannel or storage events; invalid expectation.

B. Human Gap Analysis (Items Added by Student)

To address critical blind spots inherent in generative AI models, the student supplemented **3 human items**, expanding the suite to **73 items**:

Item ID	Screen	IA	Category	Check & Expected Result	Category Missed	Why the AI Missed It (Student Analysis)
`H-01`	Cart	IA-01	Touch targets (WCAG 2.5.5)	Verify touch target sizes for interactive elements (Delete "Xóa" button, Navbar Cart link) measure $\geq 44 \times 44$ px. Expected: $\geq 44 \times 44$ px. *Observed: FAIL (25×24 px and 57×24 px).*	Mobile Accessibility / Touch Target	Generative LLMs lack physical rendering contexts and DevTools viewport metrics, habitually overlooking WCAG touch target dimensions unless explicitly prompted with CSS computed styles.
`H-02`	Cart + Checkout	IA-03	End-to-End Keyboard Flow	Verify user can complete the entire purchasing flow (remove items, proceed to checkout, apply coupon, place order) purely using `Tab`, `Shift+Tab`, `Space`, `Enter` without mouse. Expected: 100% accessible. *Observed: PASS.*	Functional Accessibility (WCAG 2.1.1)	AI evaluates keyboard focus linearly on individual components (Tab sequence per page) but fails to synthesize end-to-end multi-page keyboard operability across transactional steps.
`H-05`	Cart + Checkout	IA-01	Localization (vi-VN standards)	Verify monetary values adhere strictly to Vietnamese localization syntax: thousands dot separator `.` (`28.000.000 ₫`) rather than US comma `,` (`28,000,000 ₫`). Expected: Dot separator. *Observed: FAIL.*	Deep Regional Localization	Generative models exhibit heavy US-centric training bias, defaulting to US number formatting (`1,000.00`) and conflating internal system consistency with national typographic standards.

2.4 Execution Results Breakdown

All 73 items were rigorously executed against the running SUT (`localhost:5173`) across all 3 platforms:

Interface Aspect (IA)	Items Count	Chrome (Blink) Passed / Failed	Firefox (Gecko) Passed / Failed	Safari (WebKit) Passed / Failed	Discovered Bugs
IA-01: General UI Standards	23	17 Passed / 6 Failed	17 Passed / 6 Failed	17 Passed / 6 Failed	10 bugs (#117–#120, #123–#126, #137, #138)
IA-02: Forms & Validation	19	14 Passed / 5 Failed	14 Passed / 5 Failed	14 Passed / 5 Failed	6 bugs (#121, #127–#131)
IA-03: Navigation & Routes	19	16 Passed / 3 Failed	16 Passed / 3 Failed	16 Passed / 3 Failed	3 bugs (#132–#134)
IA-04: Feedback & State	12	4 Passed / 8 Failed	4 Passed / 8 Failed	4 Passed / 8 Failed	3 bugs (#122, #135, #136)
Total	**73 Items**	**51 Passed / 22 Failed**	**51 Passed / 22 Failed**	**51 Passed / 22 Failed**	**22 GUI Bugs**

3. Task 2: Usability Evaluation Report (40 Points)

3.1 Evaluation Plan & Objectives

- **Target Flow:** `Product Selection → Add to Cart → Apply Coupon → Checkout`.
- **Methodology:** 1-on-1 moderated assessment testing using the Concurrent Think-Aloud (CTA) protocol. The moderator acted as a neutral observer, interjecting only when participants remained silent for > 15 seconds or deadlocked for > 2 minutes.
- **Realistic Scenario Prompt:**

> *"Bạn đang cần mua một món đồ công nghệ có giá từ 300.000 ₫ trở lên phục vụ học tập, làm việc trên website EShop. Cửa hàng vừa tặng bạn một mã giảm giá là `SAVE10`. Bạn hãy tìm một sản phẩm ưng ý thỏa điều kiện trên, áp dụng mã giảm giá này và tiến hành hoàn tất việc đặt hàng."*

- **State Reset Protocol:** Between each session, local storage (`cart`) was cleared and SQLite database tables (`orders`, `coupon_usage`) were purged.
- **Benchmark Duration:** Calibrated to ≤ 150 **seconds** based on the unassisted pilot run (142s).

3.2 Participant Demographics (7 Real Users + 1 Pilot)

All participants are real individuals completely independent of the course and cohort. Contact details are maintained in the secure, gitignored `usability/participants.md` file with the middle 4 digits masked.

ID	Role / Background	Age	Occupation	E-Commerce Experience	Contact Info (Masked)	Session Date
P00	Pilot Evaluator	23	Office Administrator	Frequent (Shopee, Lazada)	`0903***124`	03/11/2026
P01	Non-IT End User	21	Marketing Student	Moderate	`0912***456`	04/11/2026

ID	Role / Background	Age	Occupation	E-Commerce Experience	Contact Info (Masked)	Session Date
P02	Non-IT End User	25	Graphic Designer	Frequent	0988***789	04/11/2026
P03	Non-IT End User	48	High School Teacher	Occasional	0905***331	05/11/2026
P04	Non-IT End User	22	Finance Associate	Frequent	0971***654	05/11/2026
P05	Non-IT End User	30	Sales Representative	Frequent	0934***890	06/11/2026
P06	Non-IT End User	52	Small Business Owner	Rare	0918***223	06/11/2026
P07	Non-IT End User	20	Pharmacy Student	Moderate	0945***908	07/11/2026

3.3 Quantitative Evaluation: System Usability Scale (SUS)

Scores were calculated using the canonical Brooke (1996) formula:

$$\text{SUS} = 2.5 \times \left[\sum_{i \in \text{odd}} (Q_i - 1) + \sum_{i \in \text{even}} (5 - Q_i) \right]$$

Participant	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	SUS Score	Adjective Rating
P01	4	2	4	1	4	2	4	2	4	2	**77.5**	Good (Above Average)
P02	4	3	3	2	4	2	4	3	3	2	**65.0**	Marginal
P03	5	2	5	1	4	2	5	2	4	1	**87.5**	Excellent
P04	5	1	5	1	5	1	5	1	5	1	**100.0**	Best Imaginable
P05	3	4	3	3	3	3	3	4	2	3	**42.5**	Poor (Frustrated by deadlock)
P06	4	2	4	2	4	2	5	2	4	2	**77.5**	Good (Above Average)
P07	5	2	4	1	5	2	5	1	4	2	**87.5**	Excellent
Mean ($\bar{x}$)	-	-	-	-	-	-	-	-	-	-	**76.79 / 100**	**Grade B / Good (+8.79 vs 68)**
Median	-	-	-	-	-	-	-	-	-	-	**77.50 / 100**	
Std Dev (s)	-	-	-	-	-	-	-	-	-	-	**18.64**	

3.4 Task Performance & Probe Ratings

- **Task Completion Rate: 100.0%** (85.7% unassisted `success`, 14.3% `assisted`).
- **Mean Duration: 140.9 seconds** (Range: 105s – 195s; 71.4% completed within ≤ 150s benchmark).
- **Post-Task Probe Dimension Ratings (1–5 Likert Scale):**
    1. **Speed & Flow: 4.43 / 5.0** (Fast, reactive single-page routing).
    2. **Trust & Security: 4.43 / 5.0** (Clear final order confirmation).
    3. **Interface Clarity: 3.71 / 5.0** (Clean product cards; subdued coupon input box).
    4. **Error Recovery: 3.29 / 5.0** (Lowest dimension: vague rejection messages when coupon fails).

3.5 Usability Findings & Criticality Prioritization

$$\text{Criticality} = \text{Severity } (1\text{–}4) \times \frac{\text{Affected Participants}}{n = 7}$$

*Small-Sample Override Rule applied: Any Major severity finding affecting ≥ 2 participants is elevated to **Fix First**.*

Finding ID	Usability Friction / Problem	Type	Severity	Affected ($n = 7$)	Probability	Criticality	Priority Band	GitHub Issue
F-01	Cryptic error message on coupon condition failure causing checkout deadlock	Genuine Bug	**Major (3)**	`[P03, P05]` (2)	0.29	0.86	**Fix First (Major Override)**	#159
F-03	Missing success confirmation toast after applying coupon code	Usability Issue	**Minor (2)**	`[P01, P07]` (2)	0.29	0.57	**Medium**	Design Fix
F-04	Low visual prominence of coupon input in Cart drawer/page	Usability Issue	**Minor (2)**	`[P01, P02]` (2)	0.29	0.57	**Medium**	Styling Fix
F-02	Coupon input field does not submit on keyboard `Enter` key press	Genuine Bug	**Minor (2)**	`[P01]` (1)	0.14	0.29	**Low**	#160
F-05	Ambiguous dual CTA buttons on	Usability Issue	**Cosmetic (1)**	`[P06]` (1)	0.14	0.14	**Low**	UX Hierarchy

Finding ID	Usability Friction / Problem	Type	Severity	Affected ($n = 7$)	Probability	Criticality	Priority Band	GitHub Issue
	slide-out cart drawer							

4. Task 3: Cross-Platform Compatibility Report (20 Points)

4.1 Test Environment Matrix

Platform ID	Browser Engine	Browser Name & Version	Operating System	Resolution / Viewport	Device Class
Platform 1	Blink	Google Chrome v124.0	Linux Ubuntu 22.04 LTS / Windows 11	1920 × 1080	Desktop Workstation
Platform 2	Gecko	Mozilla Firefox v125.0	Linux Ubuntu 22.04 LTS / Windows 11	1920 × 1080	Desktop Workstation
Platform 3	WebKit	Apple Safari v17.4 (Playwright WebKit / BrowserStack)	macOS Sonoma / iOS Emulation	1920 × 1080 / 390 × 844	Desktop & Mobile Viewport

4.2 Cross-Platform Compatibility Findings Matrix

All 22 failed checklist items were cross-executed across Chrome, Firefox, and Safari. The root cause analysis reveals uniform client-side rendering engine behavior due to shared React components and Tailwind CSS primitives:

Screen & Item ID	GitHub Issue	Chrome	Firefox	Safari	Root Cause Analysis & Engine Discrepancies
`CART-IA01-04`	#117	Failed	Failed	Failed	White text on `bg-green-500` (#22c55e) yields contrast ratio of 2.44:1, failing WCAG AA (4.5:1) uniformly across Blink, Gecko, and WebKit DevTools.
`CART-IA01-05`	#118	Failed	Failed	Failed	Inconsistent currency formatting: Home renders "VND" while Cart renders "₫". Universal across all engines.
`CART-IA01-06`	#119	Failed	Failed	Failed	Primary CTA color mismatch: Cart uses `bg-green-500` (#22c55e) while Checkout uses `bg-green-600` (#16a34a).
`CART-IA01-09`	#120	Failed	Failed	Failed	Capitalization inconsistency: Page header renders Title Case "Giỏ Hàng" while Navbar link renders Sentence case "Giỏ hàng".
`CHCK-IA01-05`	#123	Failed	Failed	Failed	White text on `bg-orange-500` (#f97316) yields contrast ratio of 2.8:1 – 2.94:1, violating WCAG AA minimums across all platforms.

Screen & Item ID	GitHub Issue	Chrome	Firefox	Safari	Root Cause Analysis & Engine Discrepancies
CHCK-IA01-07	#124	Failed	Failed	Failed	Raw unformatted integer `28000000` rendered in total payment input field without thousands separators.
CHCK-IA01-10	#125	Failed	Failed	Failed	Semantic label duplication: Screen renders "Tổng tiền thanh toán (VND):" above "Tổng thanh toán: ... ₫".
CHCK-IA01-11	#126	Failed	Failed	Failed	Casing mismatch: Label "Mã Giảm Giá" (Title Case) vs "Sản phẩm:" (Sentence case).
H-01	#137	Failed	Failed	Failed	Touch target sizes violate WCAG 2.5.5 ($\ge 44 \times 44$px): Delete button measures 25×24px; Navbar cart link measures 57×24px.
H-05	#138	Failed	Failed	Failed	Currency localization uses US comma `28,000,000 ₫` instead of Vietnamese national standard dot `28.000.000 ₫`.
CART-IA02-01	#121	Failed	Failed	Failed	Deleting a product item unmounts DOM node without managing focus, resetting `document.activeElement` to `<body>`.
CHCK-IA02-03	#127	Failed	Failed	Failed	**Critical Blocker:** Total payment amount is an editable input, allowing arbitrary negative, zero, decimal, and overflow inputs.
CHCK-IA02-04	#128	Failed	Failed	Failed	Total payment input accepts cursor focus and changes values on keyboard Arrow Up/Down or mouse wheel scrolling.
CHCK-IA02-10	#129	Failed	Failed	Failed	Data desynchronization: Applying `SAVE10` updates bottom summary label to 25,200,000 ₫ but leaves top input at 28000000.
CHCK-IA02-14	#130	Failed	Failed	Failed	Coupon input lacks `<form>` container and `onKeyDown` handler; pressing `Enter` fails to submit coupon.
CHCK-IA02-17	#131	Failed	Failed	Failed	`<label>` element lacks `htmlFor` attribute linking to input `id`; clicking label text fails to focus input.
CHCK-IA03-04	#132	Failed	Failed	Failed	Missing empty cart route guard: Directly typing `/checkout` renders checkout form with 0 items.
CHCK-IA03-05	#133	Failed	Failed	Failed	Missing authentication guard: Unauthenticated users directly navigating to `/checkout` access checkout form.
CHCK-IA03-07	#134	Failed	Failed	Failed	Browser Back button after order placement navigates back to `/cart` containing un-cleared items.
CART-IA04-03	#122	Failed	Failed	Failed	Shopping cart stored solely in React RAM state; pressing F5 reload completely clears cart.
CHCK-IA04-03	#135	Failed	Failed	Failed	Backend discount arithmetic defect: `SAVE10` discount calculates incorrect savings and total amounts.

Screen & Item ID	GitHub Issue	Chrome	Firefox	Safari	Root Cause Analysis & Engine Discrepancies
CHCK-IA04-07	#136	Failed	Failed	Failed	Missing `clearCart()` call on successful order: Returning to `/cart` still displays purchased products.

4.3 Mandatory Identity Overlay Audit

Every submitted screenshot in `evidence/screenshots/` and `evidence/usability/` incorporates:

1. Browser viewport headers and OS desktop indicators.
2. SUT running at `http://localhost:5173` or `http://localhost:3000`.
3. Visible identity watermark: **`quockinguyen.lv@gmail.com`** (Student ID: **`23120138`**).

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5. Task 4: Agent Skills & Automation Summary (10 Points)

5.1 Developed Agent Skills

```
skills/
├── gui-checklist-coach/
│   ├── SKILL.md                          # 8-phase workflow instructions &
guardrails
│   └── scripts/
│       ├── generate_checklist_excel.py   # Multi-platform openpyxl workbook
generator
│       └── create_github_issues.py       # Pilot & batch GitHub Issue
automation
└── usability-evaluation-coach/
    ├── SKILL.md                          # 10-phase usability evaluation guide
    └── scripts/
        └── score_usability.py            # Automated SUS, performance &
criticality scoring
```

1. **`gui-checklist-coach`:** Automates requirement scoping, canonical taxonomy seeding, binary testability linting, multi-platform Excel generation (`checklist.xlsx`), and GitHub Issues batch creation.
2. **`usability-evaluation-coach`:** Guides moderated think-aloud test planning, scenario linting, instrument generation, participant note structuring, automated Brooke SUS computation, and Criticality calculation with Small-Sample Major Override rules.

5.2 Demonstration Videos (YouTube Links)

Skill Name	Demonstrated Capabilities	YouTube Demonstration Link
Agent Skills Demonstration (`gui-checklist-coach` & `usability-evaluation-coach`)	End-to-end demonstration: GUI checklist scoping, taxonomy alignment, binary review, Excel generation, Pilot & Batch GitHub issues, and moderated usability evaluation with Think-Aloud protocol, automated Brooke SUS calculation, and Criticality override ranking	https://www.youtube.com/watch?v=hHmWuolx5Dk

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6. References

- ISTQB Foundation Level Syllabus v4.0 (2023).
- Course Lecture S13: *GUI Testing & Usability Testing*, CSC13003, FIT@HCMUS (2026).
- Brooke, J. (1996). *SUS: A 'quick and dirty' usability scale*.
- W3C Web Content Accessibility Guidelines (WCAG) 2.1 / 2.2.
- Kharbach, M. (2026). *AI Use Policy Templates for Higher Education*.